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Modelling with Sequences and Series

Read the story for one word: “the same amount” means arithmetic, “the same percentage” means geometric. Then it is Topic 5 or Topic 6.

THE TOPIC ON ONE MAP

Modelling with Sequences and Series

01 – Arithmetic models

same amount year n total to date

02 – Geometric models

same % $r = 1 \pm \frac{p}{100}$ logs for n

03 – Two models, limits

model A vs B limit = S_{∞} $N = ka^t$

TRANSLATE THE STORY

“increases by 5 each year” $\rightarrow d = 5$ “increases by 2% each year”
 $\rightarrow r = 1.02$ “decreases by 8%” $\rightarrow r = 0.92$

year 1 is the first term · “in year n ” = a term · “total” = a sum · “limit to the total”
 $= S_{\infty}$

01 Arithmetic Models: the Same Amount Each Time

The first year (week, row, hit) is a ; the fixed change is d . “How much in week 10” is u_{10} ; “how much altogether” is S_n . If the question gives a total, solve the quadratic for n .

WHEN YOU SEE $\rightarrow$ YOU DO

“£5 in week 1, £5.25 in week 2, £5.50 in week 3”

$\rightarrow$ arithmetic: $a = 5, d = 0.25$

“state with a reason whether arithmetic or geometric”

$\rightarrow$ constant *difference* $\rightarrow$ arithmetic

“the number of weeks to save £350”

$\rightarrow S_n \geq 350$: solve, round up

“the additional distance each hit”

$\rightarrow$ each hit is a term; total depth is the sum

WORKED MODEL · JUNE 2021

Adina saves £5, £5.25, £5.50, How many weeks to save £350?

1 $\frac{n}{2}\{10 + 0.25(n - 1)\} \geq 350$

sum

2 $n^2 + 39n - 2800 \geq 0 \Rightarrow n \geq 36.9$

quadratic

ANSWER 37 weeks

IDEA 01

Practise: Arithmetic Models: the Same Amount Each Time

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Salary rises

P2 · TOPIC 8 3 marks

A salary is £24 000 in year 1 and rises by £1500 each year. Find the salary in year 10 and the total earned over 10 years.

SELECTED PRACTICE

Q2**PRACTICE**

Practice · Easy

P2 · TOPIC 8**8 marks**

- 1 (a)** Lauren is to start training in order to run a marathon.

Each week she will run a number of miles according to the formula

$$u_n = 4n - 1$$

where u_n is the number of miles to be run in week n .

Work out how far Lauren will run in weeks 1, 2 and 3.

(2 marks)

- (b)** Work out how far Lauren will run in her 10th week of training.

(2 marks)

- (c)** Lauren tends to train for 10 weeks.

Find the total number of miles Lauren will run across all 10 weeks of her training schedule.

(3 marks)

- (d)** Explain why the model would become unrealistic for large values of n , for example for the training schedule of an elite athlete.

(1 mark)

Q3**PRACTICE** Practice · Medium**P2 · TOPIC 8 7 marks**

- 1 (a)** Lloyd is to start training in order to run a marathon.

For the first week of training he will run a total of 2 miles.

Each subsequent week he'll increase the total number of miles run by 3 miles.

He intends training for 15 weeks.

Calculate how far Lloyd will run during his eighth week of training

(2 marks)

- (b)** Work out how much further Lloyd will run in his last week of training compared to his first.

(2 marks)

- (c)** Find the total number of miles Lloyd will run across all 15 weeks of his training schedule.

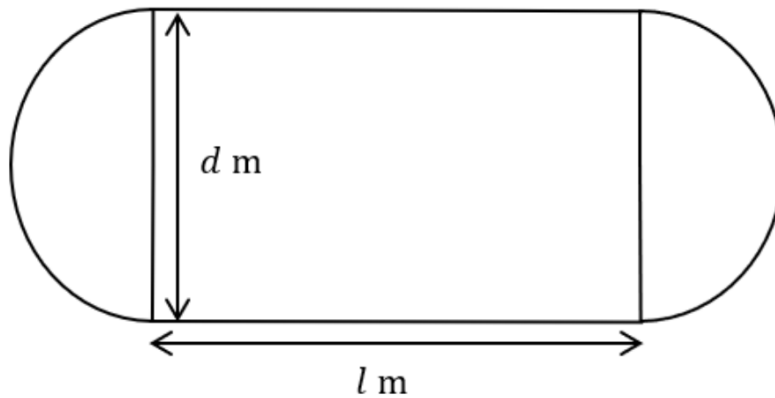
(3 marks)

Q4**PRACTICE**

Practice · Very Hard

P2 · TOPIC 8 7 marks

- 4 (a) A training track for cyclists is in the shape of a rectangle and two semi-circles as shown below. The track is also made up of several lanes.

Not
to scale

The shortest, inner lane, as shown in the diagram, has straight runs of l m, with the semi-circles at each end having a diameter of d m.

Each lane moving outwards increases the diameter by e m compared to the previous lane.

Show that the distances of each lane form an arithmetic sequence with first term $\pi d + 2l$ m and common difference πe m.

(2 marks)

- (b) There is a total of 12 lanes on the training track.

Given that l, d and e are integers and that the total distance for all 12 laps is $96(4\pi + 5)$ m, find the value of l and show that $2d + 11e = 64$.

(3 marks)

- (c) It is recommended that lanes are at least 2 m wide to allow sufficient space between cyclists in different lanes.

Find the least value of e and the associated value of d .

(2 marks)

PAST PAPERS

Q5 PAST PAPER Pearson IAL WMA12 · June 2021 · Q1

P2 · TOPIC 8 7 marks

1. Adina is saving money to buy a new computer. She saves £5 in week 1, £5.25 in week 2, £5.50 in week 3 and so on until she has enough money, in total, to buy the computer.

She decides to model her savings using either an arithmetic series or a geometric series.

Using the information given,

- (a) (i) state with a reason whether an arithmetic series or a geometric series should be used,
- (ii) write down an expression, in terms of n , for the amount, in pounds (£), saved in week n .

(3)

Given that the computer Adina wants to buy costs £350

- (b) find the number of weeks it will take for Adina to save enough money to buy the computer.

(4)

(Total 7 marks)

02 Geometric Models: the Same Percentage Each Time

Growth of $p\%$ means multiply by $1 + \frac{p}{100}$; decay means $1 - \frac{p}{100}$. The value after n steps is ar^n if you start counting at 0, ar^{n-1} if year 1 is the first term — read carefully. n in the power: logs.

WHEN YOU SEE → YOU DO

“increase by 2% each year”

→ $r = 1.02$

“the population at the start of 2032”
(2019 = first)

→ 2032 is the 14th term: ar^{13}

“three years later there were 34000”

→ $30000r^3 = 34000$

“the smallest integer N for which it exceeds 75000”

→ logs, then round up

“ $N = k\lambda^t$ ”, “half its initial value”

→ $\lambda^{5700} = \frac{1}{2}$

“in the form $N = ab^t$ ”

→ two data points: divide to find b

Find r

Write the term or sum

Take logs

Round up

WORKED MODEL • JANUARY 2020

30000 bees; 34000 three years later; $p\%$ growth. Find p .

1 $30000r^3 = 34000 \Rightarrow r = \sqrt[3]{\frac{34}{30}} = 1.0426$

three steps

ANSWER $p = 4.26$

× **TRAP** Using $r = 0.02$ for “increases by 2%”. The multiplier is 1.02.

IDEA 02

Practise: Geometric Models: the Same Percentage Each Time

My questions first, then selected practice, then real past papers.

Q6 DR ESLAM Percentage growth

P2 · TOPIC 8 3 marks

A population of 8000 grows by 5% a year. After how many complete years does it first exceed 20 000?

$r = 1.05$; take logs.

Q7 DR ESLAM Depreciation

P2 · TOPIC 8 2 marks

A car worth £20 000 loses 15% of its value each year. Find its value after 6 years.

SELECTED PRACTICE

Q8**PRACTICE** Practice · Easy**P2 · TOPIC 8 8 marks**

- 3 (a)** A ball is dropped from the top of a building and is allowed to bounce until it comes to rest. The height the ball reaches after each bounce is modelled by the geometric sequence

$$u_n = 2 \times (0.8)^{n-1}$$

[i] Write down the height the ball reaches after its first bounce and the common ratio between subsequent bounces.

[ii] Find the height the ball bounces to after its fifth bounce.

(3 marks)

- (b)** [i] Find the sum of the heights the ball reaches after 10 bounces.

[ii] Explain why the total distance travelled by the ball from the moment it first hits the ground to the moment it returns to the ground after the 10th bounce is twice your answer to part (i).

(4 marks)

- (c)** State whether the sequence $u_1, u_2, u_3, \dots$ is increasing or decreasing.

(1 mark)

Q9**PRACTICE**

Practice · Hard

P2 · TOPIC 8 9 marks

2 (a) Stephen opens a savings account with £600.

Compound interest is paid annually at a rate of 1.2%.

At the start of each new year Stephen pays another £600 into his account.

Show that at the end of two years Stephen has £1221.69 in the account

(2 marks)

(b) Show that at the end of year n , the amount of money, in pounds, Stephen will have in his account is given by

$$600(1.012 + 1.012^2 + 1.012^3 + \dots + 1.012^n)$$

(2 marks)

(c) Hence show that the total amount, in pounds, in Stephen's account after n years is

$$50\,600(1.012^n - 1)$$

(2 marks)

(d) Find the year in which the amount in Stephen's account first exceeds £10 000.

(2 marks)

(e) State one assumption that has been made about this scenario.

(1 mark)

Q10**PRACTICE**

Practice · Very Hard

P2 · TOPIC 8 5 marks

- 3 (a)** A ball is dropped from a height of A m.
It bounces to a height of x m.

Each subsequent bounce height is 25% shorter than the previous bounce.

Show that no matter how many times the ball bounces, it will not travel further than a total distance of $(A + 8x)$ m.

(4 marks)

- (b)** Write down one assumption that has been made using this model when calculating the distance.

(1 mark)

PAST PAPERS

Q11 PAST PAPER Pearson IAL WMA12 · October 2019 · Q2

P2 · TOPIC 8 6 marks

2. The adult population of a town at the start of 2019 is 25 000

A model predicts that the adult population will increase by 2% each year, so that the number of adults in the population at the start of each year following 2019 will form a geometric sequence.

- (a) Find, according to the model, the adult population of the town at the start of 2032
(3)

It is also modelled that every member of the adult population gives £5 to local charity at the start of each year.

- (b) Find, according to these models, the total amount of money that would be given to local charity by the adult population of the town from 2019 to 2032 inclusive. Give your answer to the nearest £1 000
(3)

(Total 6 marks)

Q12**PAST PAPER**

Pearson IAL WMA12 · January 2020 · Q5

P2 · TOPIC 8**8 marks**

5. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A colony of bees is being studied.

The number of bees in the colony at the start of the study was 30 000

Three years after the start of the study, the number of bees in the colony is 34 000

A model predicts that the number of bees in the colony will increase by $p\%$ each year, so that the number of bees in the colony at the end of each year of study forms a geometric sequence.

Assuming the model,

(a) find the value of p , giving your answer to 2 decimal places.

(3)

According to the model, at the end of N years of study the number of bees in the colony exceeds 75 000

(b) Find, showing all steps in your working, the smallest integer value of N .

(5)

(Total 8 marks)

Q13 PAST PAPER Pearson IAL WMA12 · June 2022 · Q9

P2 · TOPIC 8 9 marks

9. A scientist is using carbon-14 dating to determine the age of some wooden items.

The equation for carbon-14 dating an item is given by

$$N = k\lambda^t$$

where

- N grams is the amount of carbon-14 **currently** present in the item
- k grams was the **initial** amount of carbon-14 present in the item
- t is the number of years since the item was made
- λ is a constant, with $0 < \lambda < 1$

- (a) Sketch the graph of N against t for $k = 1$

(2)

Given that it takes 5700 years for the amount of carbon-14 to reduce to half its initial value,

- (b) show that the value of the constant λ is 0.999878 to 6 decimal places.

(2)

Given that Item A

- is known to have had 15 grams of carbon-14 present initially
- is thought to be 3250 years old

- (c) calculate, to 3 significant figures, how much carbon-14 the equation predicts is currently in Item A .

(2)

Item B is known to have initially had 25 grams of carbon-14 present, but only 18 grams now remain.

- (d) Use algebra to calculate the age of Item B to the nearest 100 years.

(3)

(Total 9 marks)

Q14**PAST PAPER**

Pearson IAL WMA12 · June 2024 · Q10

P2 · TOPIC 8**8 marks****10.**

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

The number of dormice and the number of voles on an island are being monitored.

Initially there are 2000 dormice on the island.

A model predicts that the number of dormice will increase by 3% each year, so that the numbers of dormice on the island at the end of each year form a geometric sequence.

- (a) Find, according to the model, the number of dormice on the island 6 years after monitoring began. Give your answer to 3 significant figures.

(2)

The number of voles on the island is being monitored over the same period of time.

Given that

- 4 years after monitoring began there were 3690 voles on the island
- 7 years after monitoring began there were 3470 voles on the island
- the number of voles on the island at the end of each year is modelled as a geometric sequence

- (b) find the equation of this model in the form

$$N = ab^t$$

where N is the number of voles, t years after monitoring began and a and b are constants. Give the value of a and the value of b to 2 significant figures.

(3)

When $t = T$, the number of dormice on the island is equal to the number of voles on the island.

- (c) Find, according to the models, the value of T , giving your answer to one decimal place.

(3)

(Total for Question 10 is 8 marks)

03 Two Models, Limits and Comparisons

Many questions compare Model A (arithmetic) with Model B (geometric) fitted to the same two facts. Find d for A and r for B from the same data, then answer each part with the right formula. A geometric model with $r < 1$ has a limit: S_{∞} .

WHEN YOU SEE → YOU DO

“model A: same amount; model B: same percentage”

→ d from $a + (n - 1)d$; r from ar^{n-1}

“the difference between the totals”

→ S_n for each model, subtract

“there is a limit to the total”

→ geometric with $r < 1$: $S_{\infty} = \frac{a}{1-r}$

“after the 20th hit the distances form a geometric sequence”

→ switch formula halfway

“state one reason why the model may not be realistic”

→ growth cannot go on for ever; values must be whole numbers

WORKED MODEL · JANUARY 2024

300 tonnes in year 1, 4000 in year 12. Model A: same amount; model B: same percentage.

1 A: $300 + 11d = 4000 \Rightarrow d = 336.4$: year 4 ≈ 1310

arithmetic

2 B: $300r^{11} = 4000 \Rightarrow r = 1.2656$: year 2 ≈ 380

geometric

ANSWER A: 1310 t, B: 380 t

IDEA 03

Practise: Two Models, Limits and Comparisons

My questions first, then selected practice, then real past papers.

Q15 **DR ESLAM** Arithmetic or geometric?

P2 · TOPIC 8 4 marks

A company sells 500 units in year 1 and 620 in year 3. Model A: arithmetic. Model B: geometric. Find, for each model, the sales in year 6.

SELECTED PRACTICE

Q16**PRACTICE**

Practice · Easy

P2 · TOPIC 8 10 marks

6 (a) Two sequences are being used to model the value of a car, n years after it was new.

At new, the car's value is £30 000.

Model 1 is an arithmetic sequence where the value of the car, u_n , at n years old, is given by the formula $u_n = 30\,000 - 5000n$.

Model 2 is a geometric sequence where the value of the car, u_n , at n years old, is given by the formula $u_n = 30\,000(0.6)^n$.

[i] Find the age of the car when Model 1 predicts its value has halved.

[ii] Find the age of the car when Model 2 predicts its value has halved.

(4 marks)

(b) Which model predicts the greater value for the car when it is 5 years old?

(3 marks)

(c) [i] State a problem with Model 1 for higher values of n .

[ii] Justify why Model 2 never predicts the value of the car to be £0.

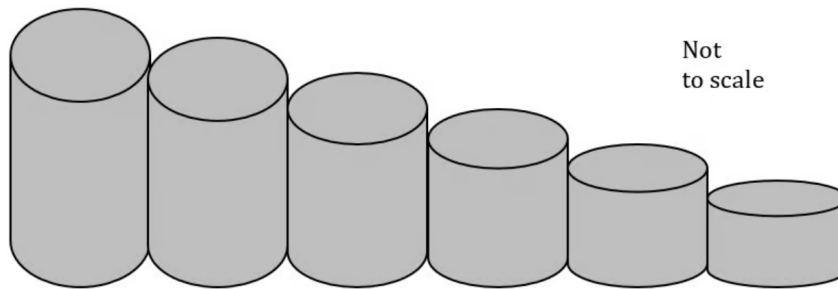
(3 marks)

Q17**PRACTICE**

Practice · Hard

P2 · TOPIC 8 5 marks

- 6 (a)** A model maker is constructing part of a model building using a series of hollow tubes, stacked next to each other as illustrated below.



Each tube is one-tenth shorter than the one to its left.

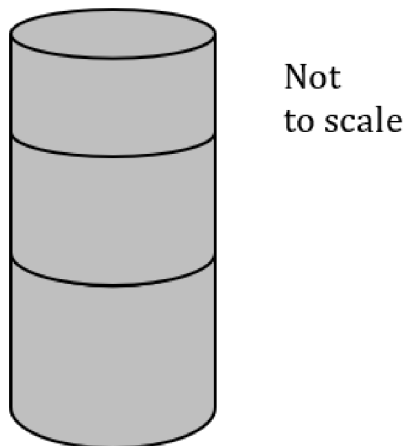
All tubes are the same width as they are all cut from one longer tube.

Show that no matter how many tubes the model maker uses, the longer tube they are cut from need not be any longer than ten times the height of the tallest tube.

(2 marks)

- (b)** In a different part of the model building, tubes are required to be stacked on top of each other, as shown below.

The tallest tube is the same length as the tallest tube above.



All the other tubes are 1 cm shorter than the tube immediately below it.

The longer tube all others are cut from is ten times the height of the tallest tube in the stack.

All of the longer tube shall be used in the stack.

If the length of the tallest tube is and there are n tubes in the stack, show that

$$n^2 - (2a + 1)n + 20a = 0.$$

(3 marks)

Q18**PRACTICE**

Practice · Very Hard

P2 · TOPIC 8 10 marks

- 5 (a)** A student is investigating a sequence of values returned from a computer program that has been reported as having a bug in it. The sequence of values is

$$3, 1, 6, \frac{1}{2}, 9, \frac{1}{4}, 12, \frac{1}{8}, 15, \frac{1}{16}, \dots$$

Explain why this sequence is neither an increasing nor a decreasing sequence.

(1 mark)

- (b)** The bug in the computer program appears to have made it output two different sequences in the same list, rather than outputting them separately.

Suggest what the two sequences could be.

(2 marks)

- (c)** To fix the bug without rewriting or searching the computer program for errors the student decides it would be easier to separate the output into one sequence using the odd numbered terms and another sequence using the even numbered terms.

Show that the odd numbered terms in the sequence form an arithmetic sequence with first term 3 and common difference 3

Find an expression for the sum of the first m terms of this sequence.

(2 marks)

- (d)** Find an expression for the sum of the first n even numbered terms of the sequence.

(2 marks)

- (e)** The computer program also outputs the sum of the first 20 terms of the sequence. (Instead of the sum of the first 20 terms for each sequence).

Find the value the computer would output, giving your answer to 4 decimal places.

(2 marks)

- (f)** Is the computer-generated sequence convergent or divergent? Give a reason for your answer.

(1 mark)

PAST PAPERS

Q19 PAST PAPER Pearson IAL WMA12 · January 2021 · Q10

P2 · TOPIC 8 11 marks

10. In this question you must show detailed reasoning.

Owen wants to train for 12 weeks in preparation for running a marathon.

During the 12-week period he will run every Sunday and every Wednesday.

- On Sunday in week 1 he will run 15 km
- On Sunday in week 12 he will run 37 km

He considers two different 12-week training plans.

In training plan *A*, he will increase the distance he runs each Sunday by the same amount.

- (a) Calculate the distance he will run on Sunday in week 5 under training plan *A*. (3)

In training plan *B*, he will increase the distance he runs each Sunday by the same percentage.

- (b) Calculate the distance he will run on Sunday in week 5 under training plan *B*.
Give your answer in km to one decimal place. (3)

Owen will also run a fixed distance, x km, each Wednesday over the 12-week period.

Given that

- x is an integer
 - the total distance that Owen will run on Sundays and Wednesdays over the 12 weeks will not exceed 360 km
- (c) (i) find the maximum value of x , if he uses training plan *A*,
(ii) find the maximum value of x , if he uses training plan *B*. (5)

Q20 **PAST PAPER** Pearson IAL WMA12 · January 2022 · Q8**P2 · TOPIC 8** **9 marks**

8. A metal post is repeatedly hit in order to drive it into the ground.

Given that

- on the 1st hit, the post is driven 100 mm into the ground
- on the 2nd hit, the post is driven an **additional** 98 mm into the ground
- on the 3rd hit, the post is driven an **additional** 96 mm into the ground
- the **additional** distances the post travels on each subsequent hit form an arithmetic sequence

(a) show that the post is driven an **additional** 62 mm into the ground with the 20th hit. (1)

(b) Find the **total distance** that the post has been driven into the ground after 20 hits. (2)

Given that for each subsequent hit after the 20th hit

- the **additional** distances the post travels form a geometric sequence with common ratio r
- on the 22nd hit, the post is driven an **additional** 60 mm into the ground

(c) find the value of r , giving your answer to 3 decimal places. (2)

After a total of N hits, the post will have been driven more than 3 m into the ground.

(d) Find, showing all steps in your working, the smallest possible value of N . (4)

(Total 9 marks)

Q21 PAST PAPER Pearson IAL WMA12 · January 2024 · Q7

P2 · TOPIC 8 9 marks

7. Wheat is grown on a farm.

- In year 1, the farm produced 300 tonnes of wheat.
- In year 12, the farm is predicted to produce 4 000 tonnes of wheat.

Model *A* assumes that the amount of wheat produced on the farm will increase by the same amount each year.

- (a) Using model *A*, find the amount of wheat produced on the farm in year 4.
Give your answer to the nearest 10 tonnes.

(3)

Model *B* assumes that the amount of wheat produced on the farm will increase by the same percentage each year.

- (b) Using model *B*, find the amount of wheat produced on the farm in year 2.
Give your answer to the nearest 10 tonnes.

(3)

- (c) Calculate, according to the two models, the difference between the total amounts of wheat predicted to be produced on the farm from year 1 to year 12 inclusive.
Give your answer to the nearest 10 tonnes.

(3)

(Total for Question 7 is 9 marks)

Q22**PAST PAPER**

Pearson IAL WMA12 · October 2024 · Q7

P2 · TOPIC 8**6 marks**

7. Jem pays money into a savings scheme, A , over a period of 300 months.

Jem pays £20 into scheme A in month 1, £20.50 in month 2, £21 in month 3 and so on, so that the amounts Jem pays each month form an arithmetic sequence.

(a) Show that Jem pays £69.50 into scheme A in month 100

(1)

(b) Find the **total** amount that Jem pays into scheme A over the period of 300 months.

(2)

Kim pays money into a different savings scheme, B , over the same period of 300 months.

In a model, the amounts Kim pays into scheme B increase by the same percentage each month, so that the amounts Kim pays each month form a geometric sequence.

Given that Kim pays

- £20 into scheme B in month 1
- £250 into scheme B in month 300

(c) use the model to calculate, to the nearest £10, the difference between the total amount paid into scheme A and the total amount paid into scheme B over the period of 300 months.

(3)

(Total for Question 7 is 6 marks)

Q23**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q9

P2 · TOPIC 8 10 marks

9. A company mines lithium.

- in year 1, the company mined 12 400 tonnes of lithium
- in year 15, the company is predicted to mine 7 500 tonnes of lithium

Model *A* assumes that the mass of lithium the company mines will decrease by the same amount each year.

According to model *A*,

- (a) find the mass of lithium the company will mine in year 5 (3)
- (b) calculate the **total** mass of lithium the company will mine from year 1 to year 15 inclusive. (2)

Model *B* assumes that the mass of lithium the company mines will decrease by the same percentage each year.

- (c) Find, according to model *B*, the mass of lithium the company will mine in year 10. Give the answer to the nearest 10 tonnes. (3)

According to model *B*, there is a limit to the **total** mass of lithium that can be mined.

- (d) Calculate the value of this limit. Give the answer to the nearest 100 tonnes. (2)

(Total for Question 9 is 10 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 Arithmetic model** Same amount = d ; term for “in year n ”, sum for “total”.
- 02 Geometric model** Same % $\rightarrow r = 1 \pm \frac{p}{100}$; logs for n .
- 03 Two models** Fit both to the same data; limit = S_{∞} .

BOTTOM LINE Same amount $\rightarrow$ arithmetic · same percentage $\rightarrow$ geometric · which year is the first term?

Answers

Modelling with Sequences and Series · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · ARITHMETIC MODELS: THE SAME AMOUNT EACH TIME

- Q1** Dr Eslam · Salary rises $24000 + 9(1500) = £37\,500$; $5(24000 + 37500) = £307\,500$
- Q2** Practice · Easy (a) 3, 7, 11 miles (b) 39 (c) $5(3 + 39) = 210$
(d) the distance keeps growing without limit
- Q3** Practice · Medium (a) $2 + 7(3) = 23$ (b) $44 - 2 = 42$ miles more
(c) $7.5(2 + 44) = 345$
- Q4** Practice · Very Hard (a) each lane: $2l + \pi d$, next lane $+\pi e$ (b)
 $6\{2(\pi d + 2l) + 11\pi e\} = 384\pi + 480 \Rightarrow l = 20$,
 $2d + 11e = 64$ (c) $e = 4$, $d = 10$
- Q5** Pearson IAL WMA12 · June 2021 · Q1 (a)(i) Arithmetic series. (a)(ii) $5 + 0.25(n-1)$.
(b) 37 weeks

IDEA 02 · GEOMETRIC MODELS: THE SAME PERCENTAGE EACH TIME

- Q6** Dr Eslam · Percentage growth $8000(1.05)^n > 20000 \Rightarrow n > 18.8$: 19 years
- Q7** Dr Eslam · Depreciation $20000(0.85)^6 \approx \text{£}7\,543$
- Q8** Practice · Easy (a)(i) 2 m, $r = 0.8$ (ii) $2(0.8)^4 \approx 0.819$ m (b)(i) $\frac{2(1-0.8^{10})}{0.2} \approx 8.93$ m (ii) each bounce goes up and comes down (c) decreasing
- Q9** Practice · Hard (a) $(607.2 + 600) \times 1.012 \approx 1221.69$ (c) $600 \times 1.012 \frac{1.012^n - 1}{0.012}$ (d) $1.012^n > 1.198 \Rightarrow n > 15.1$: year 16 (e) e.g. the rate stays at 1.2%
- Q10** Practice · Very Hard (a) $A + 2(x + 0.75x + \dots) = A + \frac{2x}{0.25} = A + 8x$ (b) e.g. it bounces for ever / no energy lost to air
- Q11** Pearson IAL WMA12 · October 2019 · Q2 (a) 32341 adults, approximately. (b) £1,997,000
- Q12** Pearson IAL WMA12 · January 2020 · Q5 (a) $p = 4.26$. (b) $N = 22$
- Q13** Pearson IAL WMA12 · June 2022 · Q9 (a) Decreasing exponential through (0, 1). (b) $\lambda = 0.999878$. (c) 10.1 g. (d) 2700 years
- Q14** Pearson IAL WMA12 · June 2024 · Q10 (a) 2390 dormice. (b) $N = 4000(0.98)^t$ to 2 s.f. constants. (c) $T = 13.9$

IDEA 03 · TWO MODELS, LIMITS AND COMPARISONS

- Q15** Dr Eslam · Arithmetic or geometric? A: $d = 60$, $500 + 5(60) = 800$; B: $r^2 = 1.24$, $500(1.24)^{2.5} \approx 856$
- Q16** Practice · Easy (a)(i) $n = 3$ (ii) $0.6^n = 0.5 \Rightarrow n \approx 1.36$ (b) Model 1: £5000, Model 2: £2333 — Model 1 (c)(i) negative after 6 years (ii) $0.6^n > 0$ for all n
- Q17** Practice · Hard (a) $h + 0.9h + 0.81h + \dots = \frac{h}{0.1} = 10h$ (b) $\frac{n}{2}\{2a - (n - 1)\} = 10a \Rightarrow n^2 - (2a + 1)n + 20a = 0$
- Q18** Practice · Very Hard (a) it goes up and down in turn (b) 3, 6, 9, ... and $1, \frac{1}{2}, \frac{1}{4}, \dots$ (c) $\frac{3m(m+1)}{2}$ (d) $2(1 - \frac{1}{2^n})$ (e) 166.9980 (f) divergent: the arithmetic terms grow without limit
- Q19** Pearson IAL WMA12 · January 2021 · Q10 (a) 23 km. (b) 20.8 km. (c)(i) $x = 4$. (c)(ii) $x = 5$
- Q20** Pearson IAL WMA12 · January 2022 · Q8 (a) Proven: 62 mm. (b) 1620 mm. (c) $r = 0.984$. (d) $N = 48$
- Q21** Pearson IAL WMA12 · January 2024 · Q7 (a) 1310 tonnes. (b) 380 tonnes. (c) 7860 tonnes
- Q22** Pearson IAL WMA12 · October 2024 · Q7 (a) Proven. (b) £28,425. (c) £1,060
- Q23** Pearson IAL WMA12 · January 2026 · Q9 (a) 11 000 tonnes (b) 149 250 tonnes (c) about 8980 tonnes (d) about 351 500 tonnes